Solution of Nonlinear Equations

Sections 5-11:

Read Chapters 5 and 6 of the textbook

Section 5

Solution of Nonlinear Equations

(Root Finding Problems)

- Definitions
- Classification of Methods
  - Analytical Solutions
  - Graphical Methods
  - Numerical Methods
    - Bracketing Methods
    - Open Methods
- Convergence Notations

Root Finding Problems

Many problems in Science and Engineering are expressed as:

Given a continuous function $f(x)$,

find the value r such that $f(r) = 0$

These problems are called root finding problems.

Roots of Equations

A number *r* that satisfies an equation is called a root of the equation.

The equation: $x^4 - 3x^3 - 7x^2 + 15x = -18$

has four roots: $-2,\ 3,\ 3,\ and\ -1$.

i.e., $x^4 - 3x^3 - 7x^2 + 15x + 18 = (x+2)(x-3)^2(x+1)$

The equation has two simple roots $(-1$ and $-2)$ and a repeated root (3) with multiplicity $= 2$.

Zeros of a Function

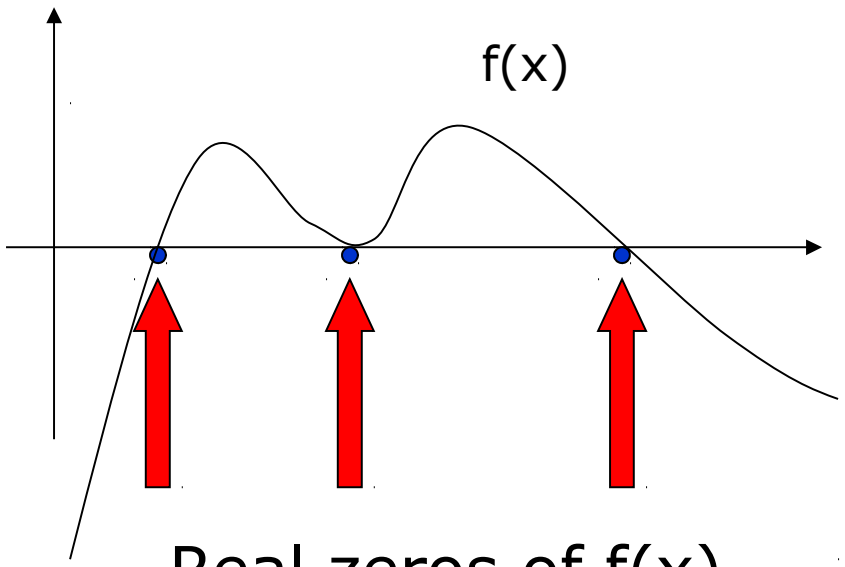
Let $f(x)$ be a real-valued function of a real variable. Any number r for which $f(r)=0$ is called a zero of the function.

Examples:

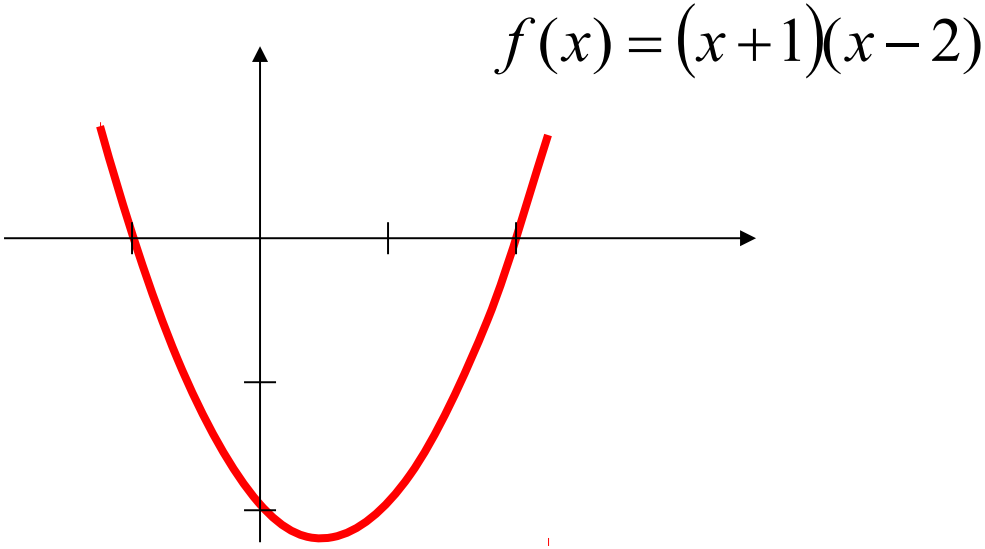
2 and 3 are zeros of the function $f(x) = (x-2)(x-3)$.

Graphical Interpretation of Zeros

- The real zeros of a function ***f(x)*** are the values of ***x*** at which the graph of the function crosses (or touches) the x-axis.



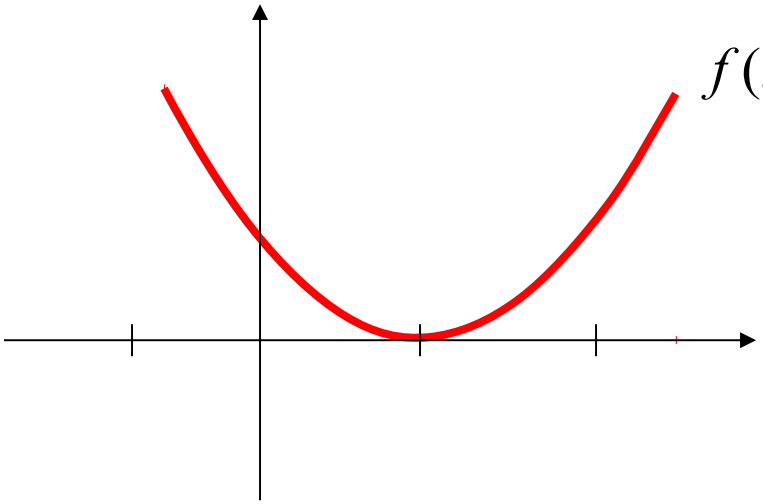
Simple Zeros



$$f(x) = (x+1)(x-2) = x^2 - x - 2$$

has two simple zeros (one at x = 2 and one at x = −1)

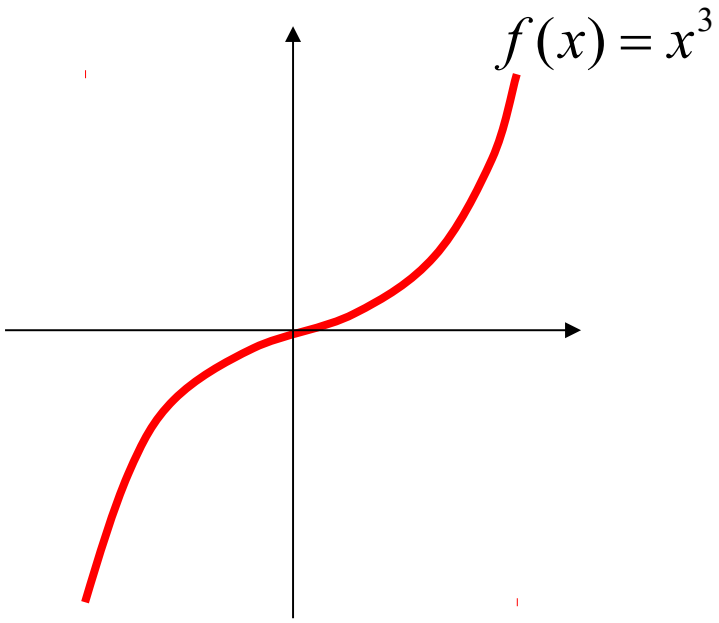
Multiple Zeros



$$f(x) = (x-1)^2 = x^2 - 2x + 1$$

has double zeros (zero with muliplicity = 2) at x = 1

Multiple Zeros



$f(x) = x^3$

has a zero with muliplicity $= 3$ at $x = 0$

Facts

- Any n^{th} order polynomial has exactly n zeros (counting real and complex zeros with their multiplicities).
- Any polynomial with an odd order has at least one real zero.
- If a function has a zero at ***x=r*** with multiplicity ***m*** then the function and its first ***(m-1)*** derivatives are zero at ***x=r*** *and the* $\boldsymbol{m}^{th}$ *derivative at* ***r*** *is not zero*.

Roots of Equations & Zeros of Function

Given the equation :

$$x^4 - 3x^3 - 7x^2 + 15x = -18$$

Move all terms to one side of the equation :

$$x^4 - 3x^3 - 7x^2 + 15x + 18 = 0$$

Define $f(x)$ as :

$$f(x) = x^4 - 3x^3 - 7x^2 + 15x + 18$$

The zeros of $f(x)$ are the same as the roots of the equation $f(x) = 0$ (Which are -2, 3, 3, and -1)

Solution Methods

Several ways to solve nonlinear equations are possible:

- Analytical Solutions
  - Possible for special equations only
- Graphical Solutions
  - Useful for providing initial guesses for other methods
- Numerical Solutions
  - Open methods
  - Bracketing methods

Analytical Methods

Analytical Solutions are available for special equations only.

Analytical solution of : $a x^2 + b x + c = 0$

$$roots = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

No analytical solution is available for : $x - e^{-x} = 0$

Graphical Methods

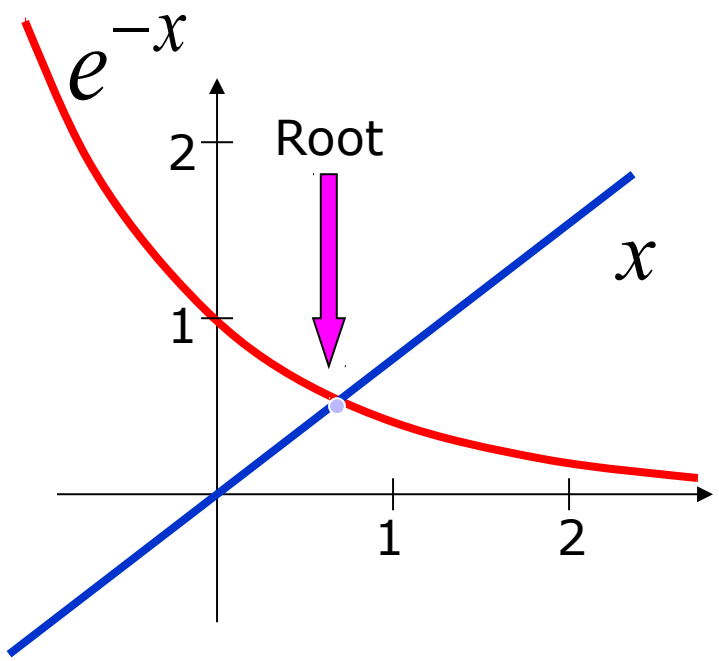
- Graphical methods are useful to provide an initial guess to be used by other methods.

$$Solve$$

$$x = e^{-x}$$

$$The\ \ root \in [0,1]$$

$$root \approx 0.6$$



Numerical Methods

Many methods are available to solve nonlinear equations:

- Bisection Method
- False position Method
- Fixed point iterations
- Newton's Method
- Secant Method

→ These will be covered in the course

- Muller's Method
- Bairstow's Method
- …….

Bracketing Methods

- In bracketing methods, the method starts with an **interval** that contains the root and a procedure is used to obtain a smaller interval containing the root.

- Examples of bracketing methods:
  - **Bisection method**
  - **False position method**

Open Methods

- In the open methods, the method starts with one or more initial guess points. In each iteration, a new guess of the root is obtained.
- Open methods are usually **more efficient** than bracketing methods.
- They **may not converge** to a root.

Bisection Method

- The Bisection Algorithm
- Examples

Introduction

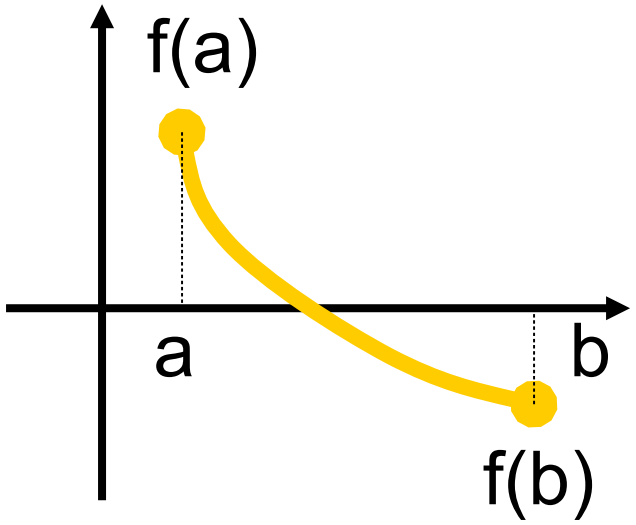
- The **Bisection method** is one of the simplest methods to find a zero of a nonlinear function.
- It is also called **interval halving** method.
- To use the Bisection method, one needs an initial interval that is known to contain a zero of the function.
- The method systematically reduces the interval. It does this by **dividing the interval into two equal** parts, performs a simple test and based on the result of the test, half of the interval is thrown away.
- The procedure is repeated until the desired interval size is obtained.

Intermediate Value Theorem

- Let f(x) be defined on the interval [a,b].

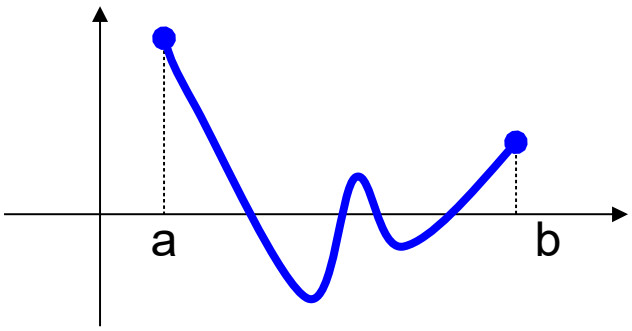
- Intermediate value theorem:

  if a function is continuous and f(a) and f(b) have different signs then the function has at least one zero in the interval [a,b].



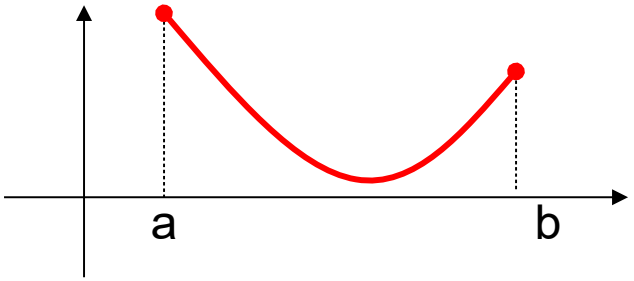
Examples

- If f(a) and f(b) have the same sign, the function may have an even number of real zeros or no real zeros in the interval [a, b].



The function has four real zeros

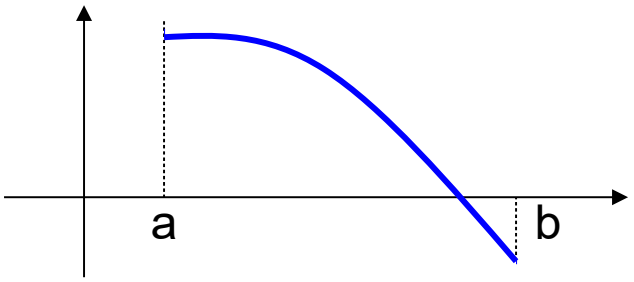
- Bisection method can not be used in these cases.



The function has no real zeros

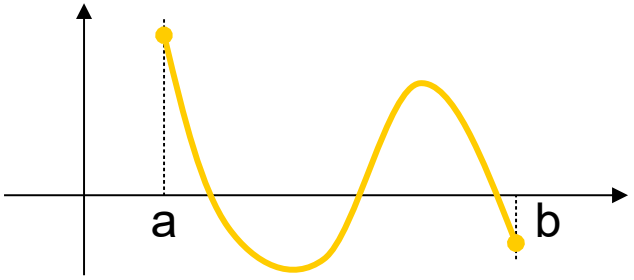
Two More Examples

- If f(a) and f(b) have different signs, the function has at least one real zero.



The function has one real zero

- Bisection method can be used to find one of the zeros.



The function has three real zeros

Bisection Method

- If the function is continuous on [a,b] and f(a) and f(b) have different signs, Bisection method obtains a new interval that is half of the current interval and the sign of the function at the end points of the interval are different.

- This allows us to repeat the Bisection procedure to further reduce the size of the interval.

Bisection Method

Assumptions:

Given an interval [a,b]

f(x) is continuous on [a,b]

f(a) and f(b) have opposite signs.

These assumptions ensure the existence of at least one zero in the interval [a,b] and the bisection method can be used to obtain a smaller interval that contains the zero.

Bisection Algorithm

Assumptions:

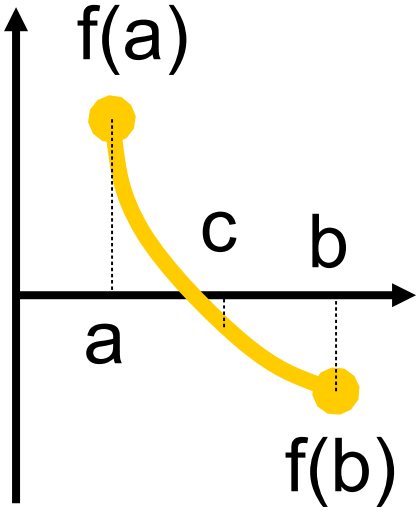
- f(x) is continuous on [a,b]
- f(a) f(b) < 0

Algorithm:

Loop

1. Compute the mid point c=(a+b)/2
2. Evaluate f(c)
3. If f(a) f(c) < 0 then new interval [a, c]
   If f(a) f(c) > 0 then new interval [c, b]

End loop



Bisection Method

Step 1: Choose lower x_l and upper x_u guesses for the root such that the function changes sign over the interval. This can be checked by ensuring that $f(x_l)f(x_u) < 0$.

Step 2: An estimate of the root x_r is determined by

$$x_r = \frac{x_l + x_u}{2}$$

Step 3: Make the following evaluations to determine in which subinterval the root lies:

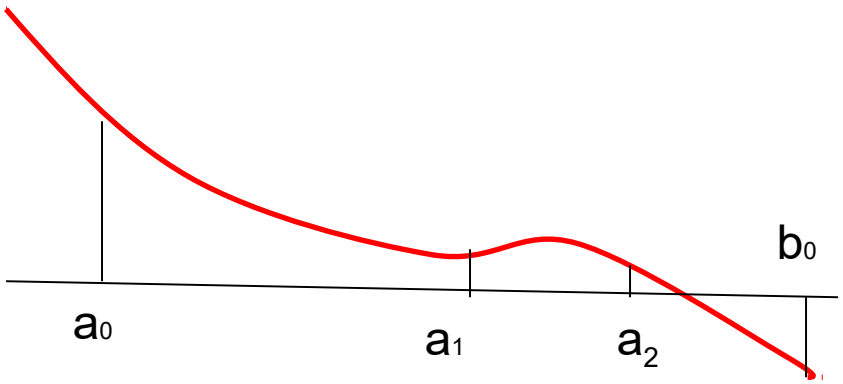
(*a*) If $f(x_l)f(x_r) < 0$, the root lies in the lower subinterval. Therefore, set $x_u = x_r$ and return to step 2.

(*b*) If $f(x_l)f(x_r) > 0$, the root lies in the upper subinterval. Therefore, set $x_l = x_r$ and return to step 2.

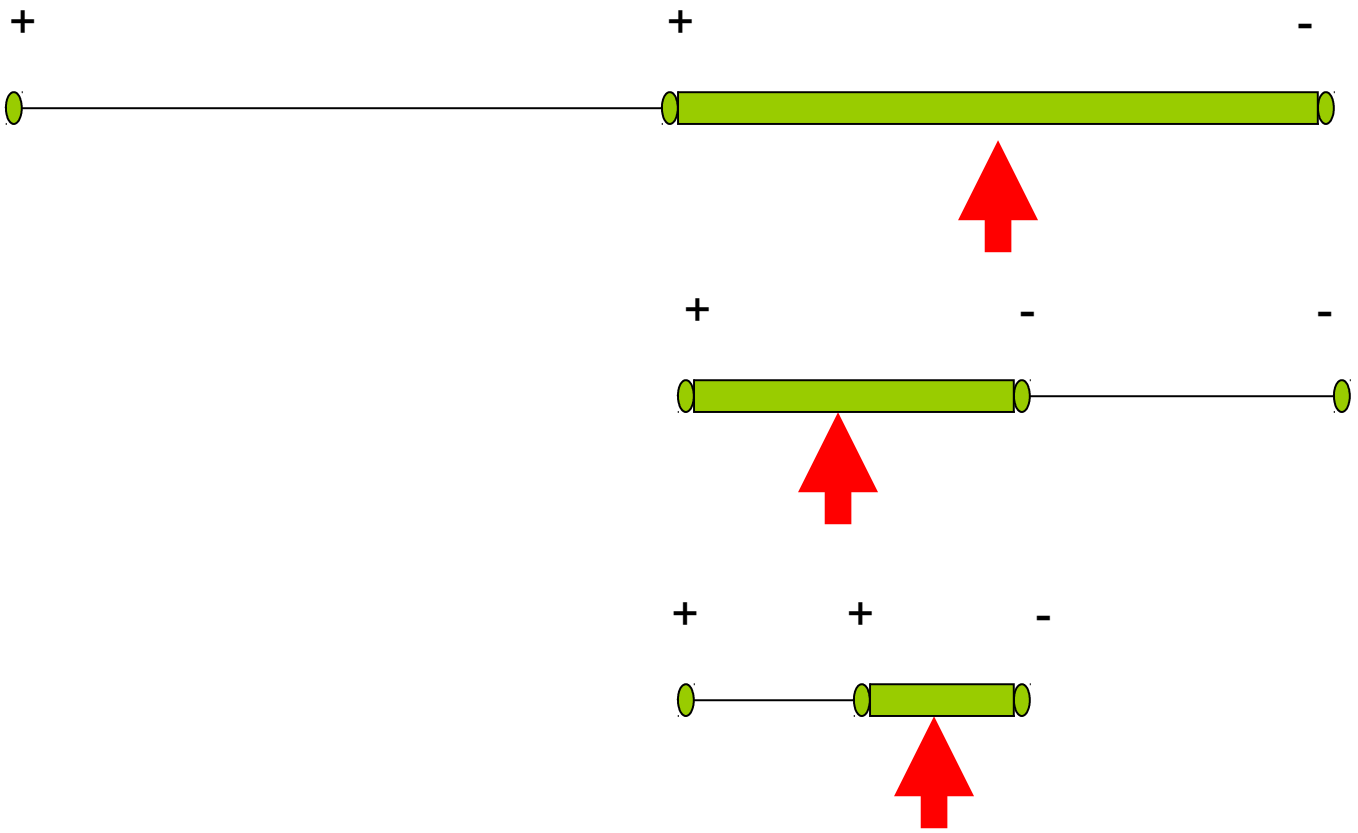
(*c*) If $f(x_l)f(x_r) = 0$, the root equals x_r; terminate the computation.

Created with

Bisection Method



Example



Best Estimate and Error Level

The best estimate of the zero of the function ***f(x)*** after the first iteration of the Bisection method is the mid point of the initial interval:

$$Estimate\ of\ the\ zero: \quad r = \frac{b+a}{2}$$

$$Error \le \frac{b-a}{2}$$ absolute error

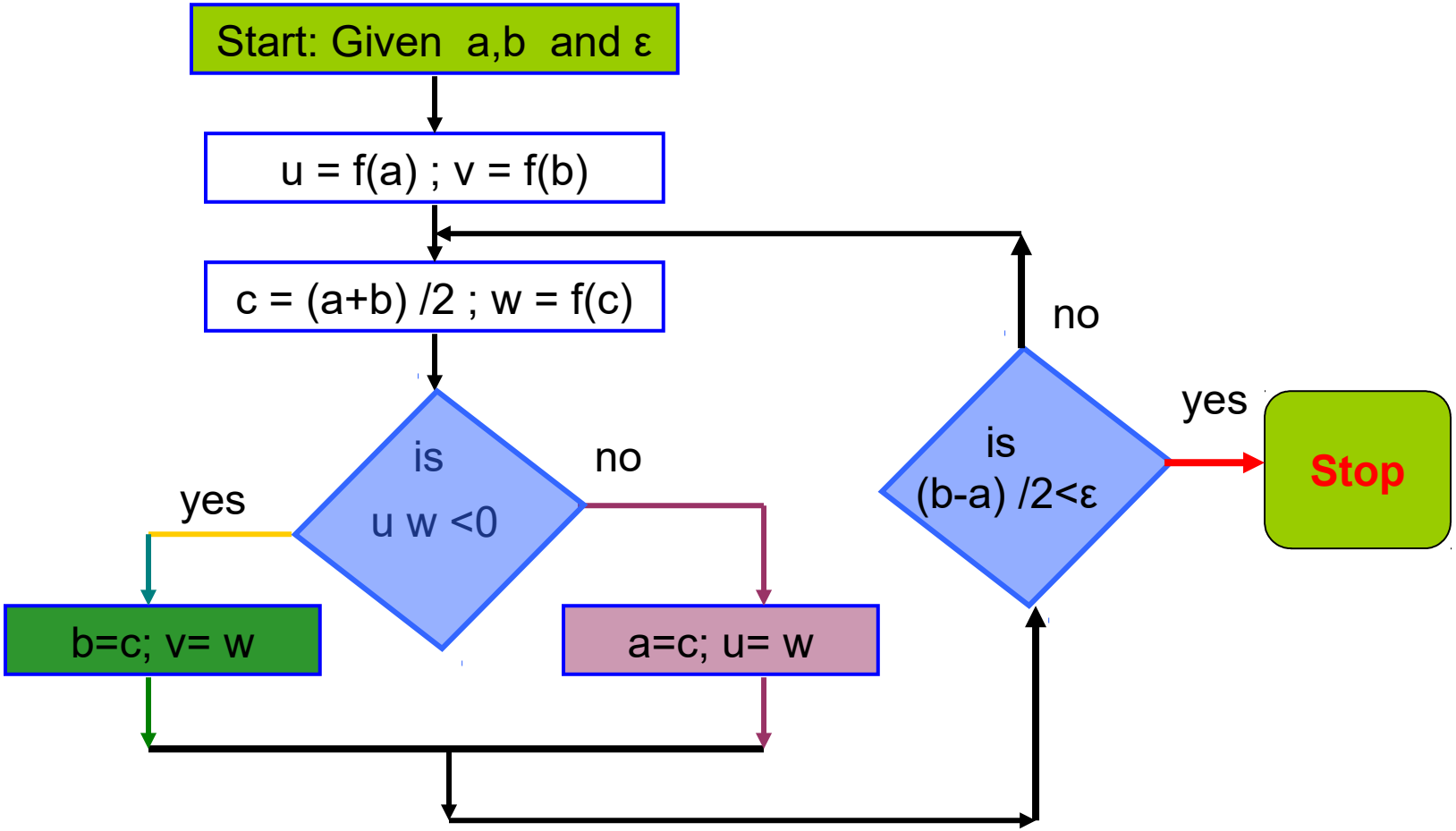
Stopping Criteria

Two common stopping criteria:

1. Stop after a fixed number of iterations
   - This condition is used as a safety mechanism to prevent infinite loops.
2. Stop when the absolute error is less than a specified value
   - abs(b-a)/2 <tolerance (Accepted Error)

How are these criteria related?

Flow Chart of Bisection Method



Other stopping criteria

The **Percent Relative Approximate Error** is:

$$\varepsilon_a(\%) = \left| \frac{x_r^{(k)} - x_r^{(k-1)}}{x_r^{(k)}} \right| \times 100 \quad < \varepsilon_s$$

Where:

- $x_r^{(k)}$ = current root approximation
- $x_r^{(k-1)}$ = previous root approximation
- ε_a = percent relative approximate error

For n significant figures, we usually choose tolarance:

$$\varepsilon_s = 0.5 \times 10^{2-n}$$

Other stopping criteria

- Function Value Criterion
  - Stop when:

$$|f(x_r)| < \varepsilon$$

  - This means the function value is sufficiently close to zero.

Example

Can you use Bisection method to find a zero of :

$f(x) = x^3 - 3x + 1$ in the interval [0,2]?

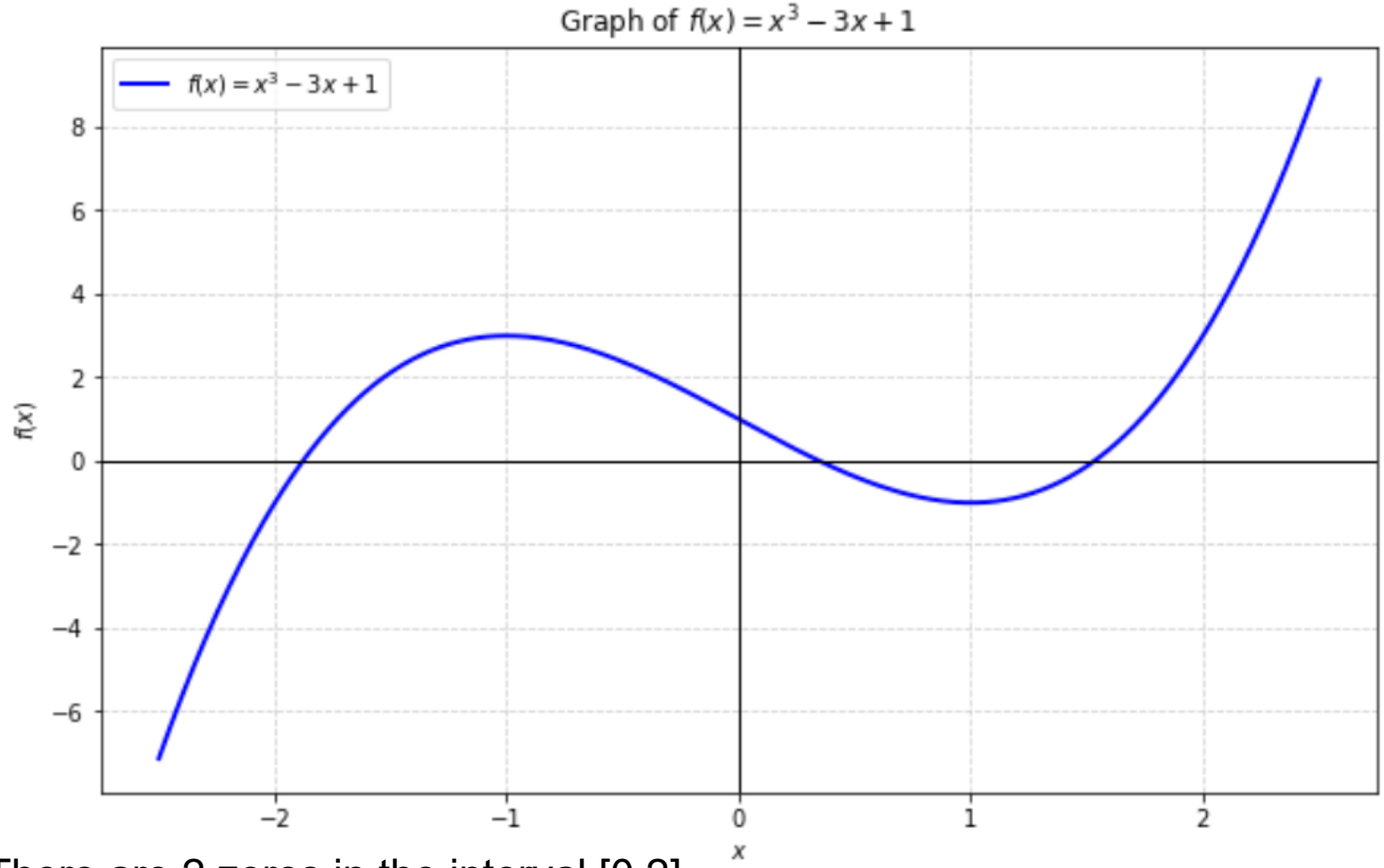
Answer:

$f(x)$ is continuous on [0,2]

and $f(0) * f(2) = (1)(3) = 3 > 0$

$\Rightarrow$ Assumptions are not satisfied

$\Rightarrow$ Bisection method can not be used



There are 2 zeros in the interval [0,2]

Example

Can you use Bisection method to find a zero of :

$f(x) = x^3 - 3x + 1$ in the interval $[0,1]$?

Answer:

$f(x)$ is continuous on $[0,1]$

and $f(0) * f(1) = (1)(-1) = -1 < 0$

$\Rightarrow$ Assumptions are satisfied

$\Rightarrow$ Bisection method can be used

Example 1 $f(x) = x^3 - 3x + 1$ in the interval :[0,1]

#	a	f(a)	b	f(b)	c= (a+b) /2	f(c)	App. Relative Error	(b-a) 2
1								
2								
3								
4								
5								

Example

$f(x) = x^3 - 3x + 1$ in the interval : $[0,1]$

#	a	*f(a)*	*b*	*f(b)*	*c= (a+b)/2*	*f(c)*	*App. Relative Error*	*(b-a) / 2*
1	**0**	**1**	**1**	**-1**	**0.5**	**-0.375**	-	**0.5**
2								
3								
4								
5								

Example

$f(x) = x^3 - 3x + 1$ in the interval $:[0,1]$

#	a	*f(a)*	*b*	*f(b)*	*c= (a+b)/2*	*f(c)*	*App. Relative Error*	*(b-a) 2*
1	0	1	1	-1	0.5	-0.375	-	0.5
2	0	1	0.5	-0.375	0.25	0.266	|0.25-.5/0.25|X 100 =100	0.25
3								
4								
5								

Example

$f(x) = x^3 - 3x + 1$ in the interval $:[0,1]$

#	a	*f(a)*	*b*	*f(b)*	*c= (a+b)/2*	*f(c)*	*App. Relative Error*	*(b-a) 2*
1	0	1	1	-1	0.5	-0.375	-	0.5
2	0	1	0.5	-0.375	0.25	0.266	|0.25-.5/0.25|X 100 =100	0.25
3	0.25	0.266	0.5	-0.375	.375	-7.23E-3	33.33 %	0.125
4								
5								

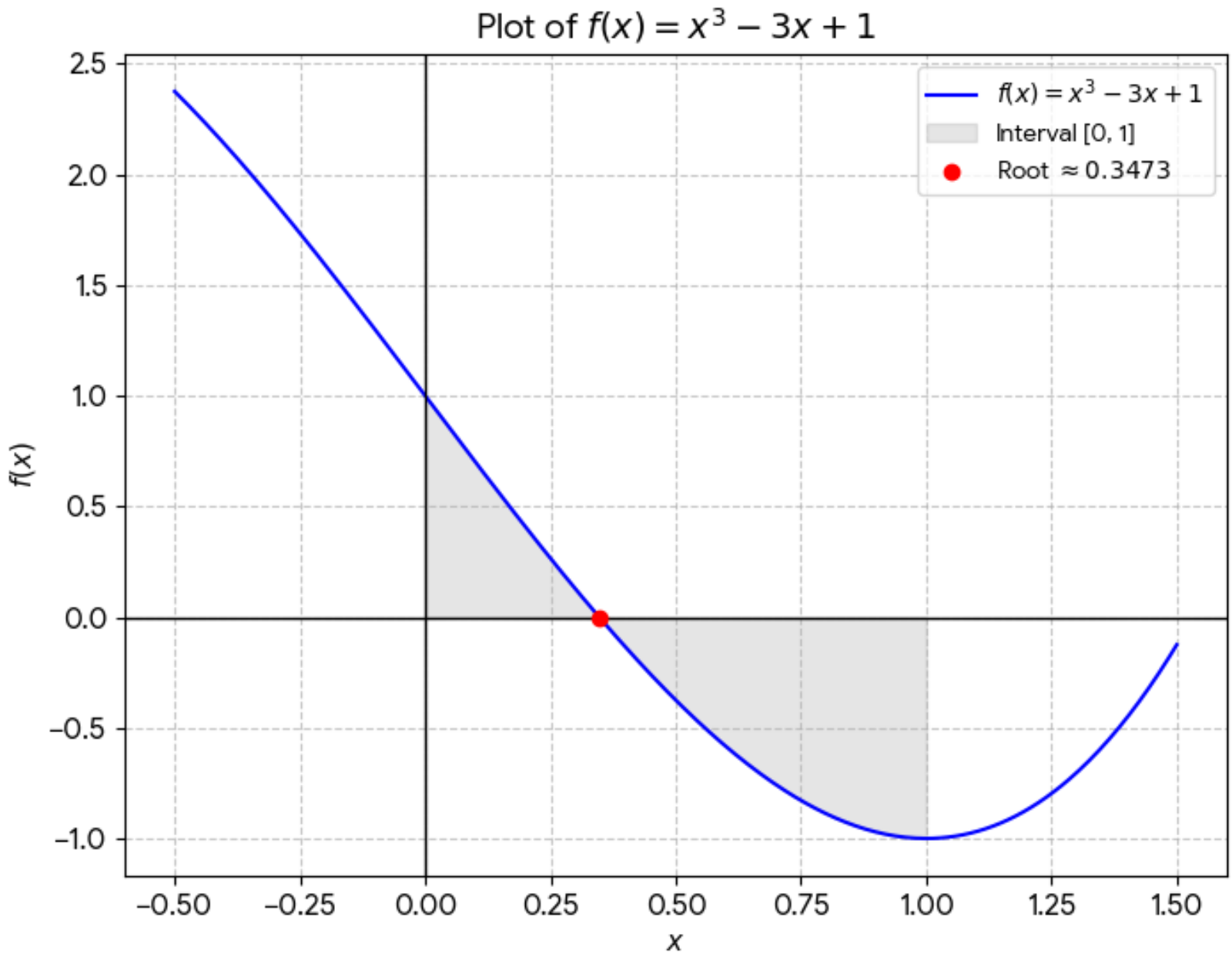
Example

$f(x) = x^3 - 3x + 1$ in the interval $:[0,1]$

#	a	*f(a)*	*b*	*f(b)*	*c= (a+b)/2*	*f(c)*	*App. Relative Error*	*(b-a) 2*
1	0	1	1	-1	0.5	-0.375	-	0.5
2	0	1	0.5	-0.375	0.25	0.266	|0.25-.5/0.25|X 100 =100	0.25
3	0.25	0.266	0.5	-0.375	.375	-7.23E-3	33.33 %	0.125
4	0.25	0.266	0.375	-7.23E-3	0.3125	9.30E-2	20%	0.0625
5	0.3125	9.30E-2	0.375	-7.23E-3	0.34375	9.37E-3	9.09 %	0.03125

After 20 iterations

Iter	a	f(a)	b	f(b)	c	f(c)	abs(b-a)/2)
1	0.0000	1.000000	1.0000	-1.000000	0.5000	-0.375000	0.50000000
2	0.0000	1.000000	0.5000	-0.375000	0.2500	0.265625	0.25000000
3	0.2500	0.265625	0.5000	-0.375000	0.3750	-0.072266	0.12500000
4	0.2500	0.265625	0.3750	-0.072266	0.3125	0.093018	0.06250000
5	0.3125	0.093018	0.3750	-0.072266	0.3438	0.009369	0.03125000
6	0.3438	0.009369	0.3750	-0.072266	0.3594	-0.031712	0.01562500
7	0.3438	0.009369	0.3594	-0.031712	0.3516	-0.011236	0.00781250
8	0.3438	0.009369	0.3516	-0.011236	0.3477	-0.000949	0.00390625
9	0.3438	0.009369	0.3477	-0.000949	0.3457	0.004206	0.00195312
10	0.3457	0.004206	0.3477	-0.000949	0.3467	0.001627	0.00097656
11	0.3467	0.001627	0.3477	-0.000949	0.3472	0.000339	0.00048828
12	0.3472	0.000339	0.3477	-0.000949	0.3474	-0.000305	0.00024414
13	0.3472	0.000339	0.3474	-0.000305	0.3473	0.000017	0.00012207
14	0.3473	0.000017	0.3474	-0.000305	0.3474	-0.000144	0.00006104
15	0.3473	0.000017	0.3474	-0.000144	0.3473	-0.000064	0.00003052
16	0.3473	0.000017	0.3473	-0.000064	0.3473	-0.000024	0.00001526
17	0.3473	0.000017	0.3473	-0.000024	0.3473	-0.000003	0.00000763
18	0.3473	0.000017	0.3473	-0.000003	0.3473	0.000007	0.00000381
19	0.3473	0.000007	0.3473	-0.000003	0.3473	0.000002	0.00000191
20	0.3473	0.000002	0.3473	-0.000003	0.3473	-0.000001	0.00000095


Plot of $f(x) = x^3 - 3x + 1$
$f(x) = x^3 - 3x + 1$
Interval [0, 1]
Root ≈ 0.3473
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
f(x)
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
x

Exercise

Find the root of the function

$$f(x) = 0.4x^2 + 2.2x + 4.7$$

using Bisection method in the interval **[5,10]**, where the **acceptance error is 5%**

Convergence Analysis

In Numerical Analysis, convergence means: **A sequence of approximations gets closer and closer to the exact solution.**

Given $f(x)$, a, b, *and* ε

How many iterations are needed such that: $|x - r| \leq \varepsilon$ where r is the zero of $f(x)$ and x is the bisection estimate (i.e., $x = c_n$)?

Convergence Notation

A sequence $x_1, x_2, ..., x_n, ...$ is said to **converge** to x if to every $\varepsilon > 0$ there exists N such that:

$$|x_n - x| < \varepsilon \quad \forall n > N$$

The order of convergence measures how fast an iterative sequence xn approaches the exact root

Convergence Notation

Suppose:

$$e_n = 0.01$$

Linear case (p = 1, C = 0.5):

$$e_{n+1} = 0.005$$

Small improvement.

Quadratic case (p = 2, C = 1):

$$e_{n+1} = (0.01)^2 = 0.0001$$

Huge improvement.

Convergence Notation

Let $x_1, x_2, \ldots,$ converge to x.

Linear Convergence : $$\frac{|x_{n+1} - x|}{|x_n - x|} \leq C$$

Quadratic Convergence : $$\frac{|x_{n+1} - x|}{|x_n - x|^2} \leq C$$

Convergence of order P : $$\frac{|x_{n+1} - x|}{|x_n - x|^p} \leq C$$

Speed of Convergence

- We can compare different methods in terms of their convergence rate.
- Quadratic convergence is faster than linear convergence.
- A method with convergence order ***q*** converges faster than a method with convergence order ***p*** if ***q>p***.

Best Estimate and Error Level

The best estimate of the zero of the function ***f(x)*** after the first iteration of the Bisection method is the mid point of the initial interval:

$$Estimate\ of\ the\ zero: \quad r = \frac{b+a}{2}$$

$$Error \leq \frac{b-a}{2}$$

Stopping Criteria

Two common stopping criteria

1. Stop after a fixed number of iterations
2. Stop when the absolute error is less than a specified value

How are these criteria related?

Absolute Error

- Initial interval = $[a, b]$
- Initial length = $L_0 = b - a$
- Required tolerance = ε

After n iterations, the interval length becomes:

$$L_n = \frac{b - a}{2^n}$$

To guarantee the error is less than or equal to ε:

$$\frac{b - a}{2^n} \leq \varepsilon$$

Absolute Error

After:

- 1 iteration → length = $\frac{b-a}{2}$
- 2 iterations → length = $\frac{b-a}{2^2}$
- n iterations → length =

$$L_n = \frac{b-a}{2^n}$$

Minimum Number of Iterations

Solve for n:

$$n \geq \log_2 \left(\frac{b-a}{\varepsilon} \right)$$

So the **minimum number of iterations** is:

$$n = \left\lceil \log_2 \left(\frac{b-a}{\varepsilon} \right) \right\rceil$$

Where:

- $\lceil \cdot \rceil$ = ceiling function (round up)

Convergence Analysis

the minimum number of iterations is:

$$n \geq \log_2\left(\frac{\text{width of initial interval}}{\text{desired error}}\right) = \log_2\left(\frac{b-a}{\varepsilon}\right)$$

Example

$a = 6,\ b = 7,\ \varepsilon = 0.0005$

How many iterations are needed such that: $|x - r| \le \varepsilon$?

$$n = \left\lceil \log_2 \left(\frac{b-a}{\varepsilon} \right) \right\rceil = 10.9658$$

$$\Rightarrow n \ge 11$$

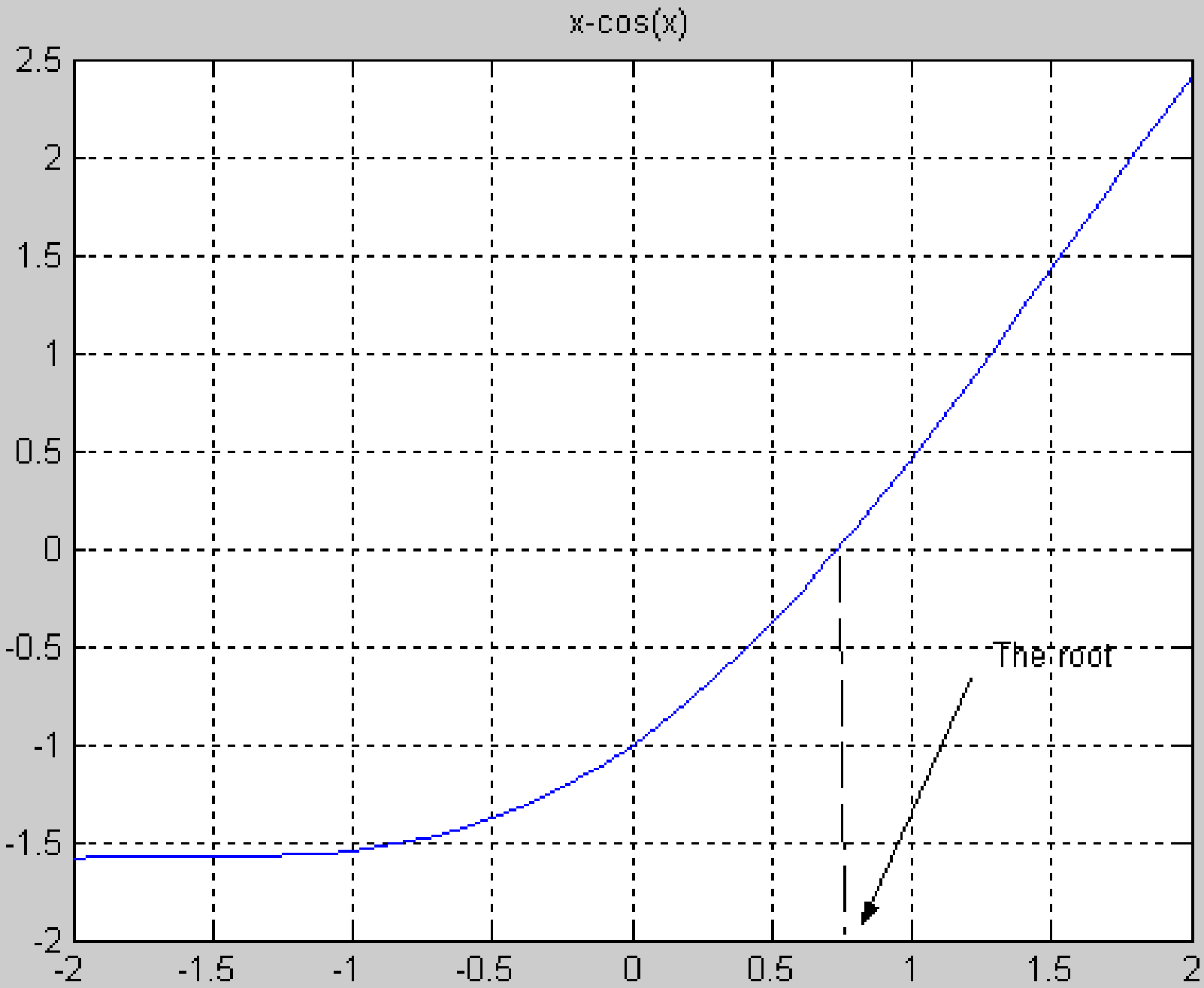
Example

- Use Bisection method to find a root of the equation f(x) = cos (x) with absolute error <0.02 (assume the initial interval [0.5, 0.9])

Question 1: What is *f (x) ?*
Question 2: Are the assumptions satisfied ?
Question 3: How many iterations are needed ?
Question 4: How to compute the new estimate ?

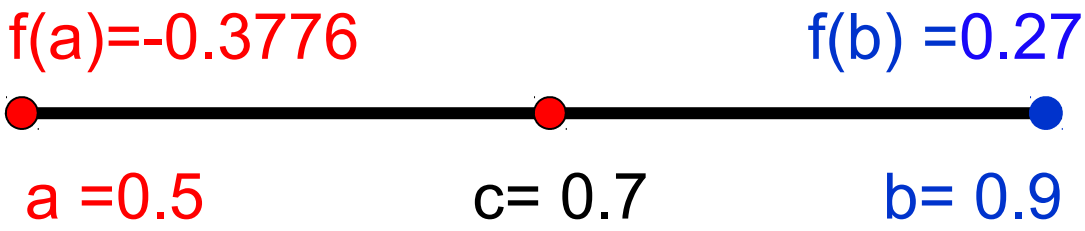
$$n \geq \log_2 \left(\frac{b-a}{\varepsilon} \right)$$

$\text{Log}_{2(}(0.9\text{-}0.5)/0.02)=4.3219 \rightarrow 5$


x-cos(x)
2.5
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2
The root

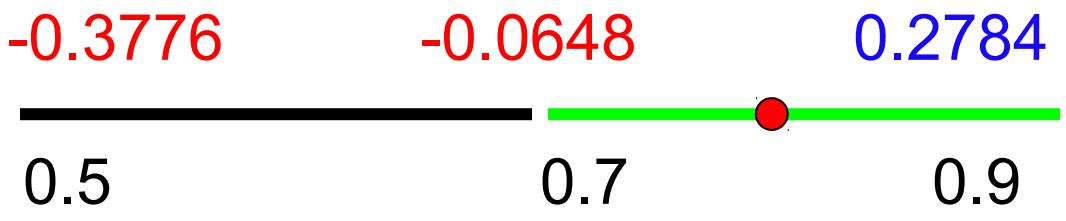
Bisection Method

Initial Interval

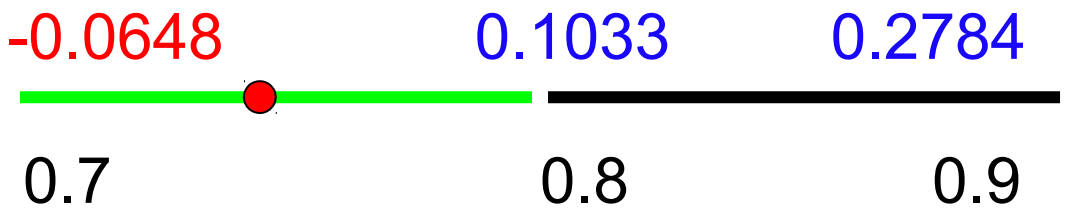


Error < 0.2

Bisection Method



Error < 0.1



Error < 0.05

Bisection Method



Error < 0.025



Error < .0125

Summary

- Initial interval containing the root: [0.5,0.9]

- After 5 iterations:
  - Interval containing the root: [0.725, 0.75]
  - Best estimate of the root is 0.7375
  - | Error | < 0.0125

Bisection Method

Advantages

- Simple and easy to implement
- One function evaluation per iteration
- The size of the interval containing the zero is reduced by 50% after each iteration
- The number of iterations can be determined a priori
- No knowledge of the derivative is needed
- The function does not have to be differentiable

Disadvantage

- Slow to converge
- Good intermediate approximations may be discarded

Selected Problems from textbook

5.1 Determine the real roots of $f(x) = -0.6x^2 + 2.4x + 5.5$:

(a) Graphically.

(b) Using the quadratic formula.

(c) Using three iterations of the bisection method to determine the highest root. Employ initial guesses of $x_l = 5$ and $x_u = 10$. Compute the estimated error ε_a and the true error ε_t after each iteration.

5.2 Determine the real root of $f(x) = 4x^3 - 6x^2 + 7x - 2.3$:

(a) Graphically.

(b) Using bisection to locate the root. Employ initial guesses of $x_l = 0$ and $x_u = 1$ and iterate until the estimated error ε_a falls below a level of $\varepsilon_s = 10\%$.

5.3 Determine the real root of $f(x) = -26 + 85x - 91x^2 + 44x^3 - 8x^4 + x^5$:

(a) Graphically.

(b) Using bisection to determine the root to $\varepsilon_s = 10\%$. Employ initial guesses of $x_l = 0.5$ and $x_u = 1.0$.

(c) Perform the same computation as in **(b)** but use the false-position method and $\varepsilon_s = 0.2\ \%$.